

Qatar-2b

R-0942

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_180323 · created 2026-07-29T13:11:33+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458200.89001 +/- 0.0006 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.066
Transit depth (Rp/Rs)^2	8.4e-06 +/- 0.0038 [%]
Orbital Inclination (inc)	87.52 +/- 1.73
Airmass coefficient 1 (a1)	2.32 +/- 0.96
Airmass coefficient 2 (a2)	-0.68 +/- 0.29
Scatter in the residuals of the lightcurve fit is	36.36 %
Best Comparison Star	#2 - [165, 463]
Optimal Method	PSF photometry
Transit Duration (day)	0.0656 +/- 0.0055

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.0 ± 0.066 vs 0.16327 (deviation 1.69σ)
Duration fit vs published	0.0656 d vs 0.0767 d (deviation 1.38σ)
Mid-time O-C	-1.0 min
Depth SNR	0.0
Residual scatter	36.36 %

Light curve

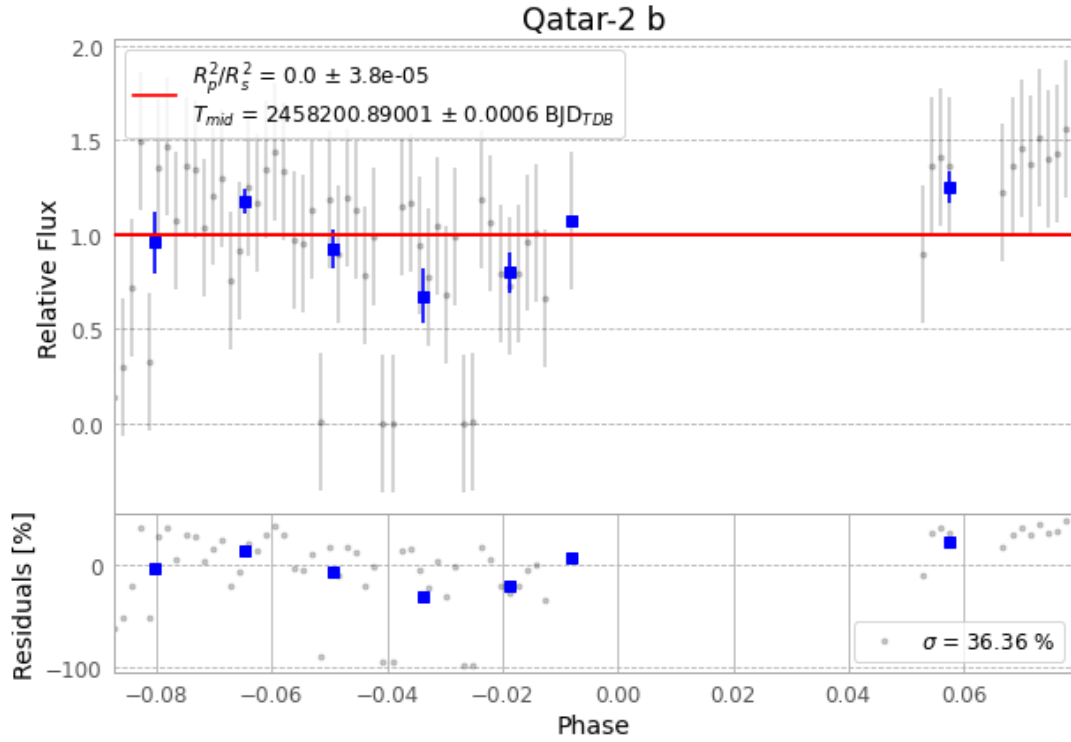


Figure 1 — FinalLightCurve_Qatar-2 b_2018-03-22.png (SHA-256 1eca6fb0ce6a64b6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [523, 404], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9508bf2d9e4573df...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

108 FITS frames (71 MB), Qatar-2180323062412.FITS → Qatar-2180323114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-180323.log (8d8b0df1ea26...), FinalLightCurve_Qatar-2 b_2018-03-22.png (1eca6fb0ce6a...), FinalLightCurve_Qatar-2 b_2018-03-22.csv (6e6bb9fe2d99...)

Compute resources

Wall time 180.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)